Team Bio

Project: AdaptivePool: A Queue-Aware Adaptive Thread Pool for a Concurrent Web Server (OSTEP Project G.5)

Yang (Selena) Zhang

- **Email:** [your Northeastern email]
- **Role (lead):** Workload generator and instrumentation. I will build the load generator that produces steady and bursty traffic with different CPU / I/O mixes, and the timestamp logging we use to measure latency.
- **Strengths for this project:** Backend development experience from my internship at a logistics company, where I built a shipment tracking and anomaly alerting platform that dealt with a steady stream of incoming events. I am comfortable with Python for scripting experiments and plotting results.
- **Project goal:** [edit] Produce clean, repeatable experiments that clearly show whether the adaptive pool helps tail latency.
- **Something I want to learn:** [edit] How to write correct multi-threaded C code with pthreads, mutexes and condition variables, and how to measure tail latency properly.

[Teammate 2]

- **Email:** [email]
- **Role (lead):** [suggested] Baseline concurrent server: master/worker thread pool, bounded buffer with condition variables, FIFO and SFF scheduling.
- **Strengths for this project:** [fill in]
- **Project goal:** [fill in]
- **Something they want to learn:** [fill in]

[Teammate 3]

- **Email:** [email]
- **Role (lead):** [suggested] AdaptivePool research extension: monitor thread, grow/shrink policy, safe worker retirement.
- **Strengths for this project:** [fill in]
- **Project goal:** [fill in]
- **Something they want to learn:** [fill in]

Shared Responsibilities

- Everyone runs one of the three existing systems for the exploration report and writes its analysis.
- Everyone logs their time in the contribution tracker after each work session.
- Experiments and the final analysis (Sprint 3) are done together, with the workload owner coordinating the runs.